

Predicting Recreational Runners' Marathon Performance Time During Their Training Preparation

Jonathan Esteve-Lanao,¹ Sebastián Del Rosso,³ Eneko Larumbe-Zabala,⁴ Claudia Cardona,^{1,5} Alberto Alcocer-Gamboa,² and Daniel A. Boullosa^{6,7}

¹All In Your Mind Training System TM, Yucatán, Mexico; ²Winhealth Medical Center, Merida, Yucatán, Mexico; ³Post-Graduate Program in Physical Education, Catholic University of Brasília, Brasília, Brazil; ⁴Clinical Research Institute, Texas Tech University Health Sciences Center, Lubbock, Texas; ⁵Health Sciences Faculty, University of the Valley of Mexico, Mérida, Yucatán, Mexico; ⁶iLOAD Solutions, Brasília, DF, Brazil; and ⁷Sport and Exercise Science, James Cook University, Townsville, Australia

Abstract

Esteve-Lanao, J, Del Rosso, S, Larumbe-Zabala, E, Cardona, C, Alcocer-Gamboa, A, and Boullosa, DA. Predicting marathon performance time throughout the training preparation in recreational runners. *J Strength Cond Res* XX(X): 000–000, 2019—The objective of this study was to predict marathon performance at different time points along the season using different speeds derived from ventilatory thresholds and running economy (RE). Sixteen recreational runners (8 women and 8 men) completed a 16-week marathon training macrocycle. Aerobic threshold (AeT), anaerobic threshold (AnT), and maximal oxygen uptake were assessed at the beginning of the season, whereas speeds eliciting training zones at AeT and AnT, and RE were evaluated at 5-time points during the season (M1–M5). Analyses of variance and hierarchical regression analyses were conducted. Training improved AeT and AnT speeds at M2 vs. M1 ($p = 0.001$) and remained significantly higher at M3, M4, and M5 ($p = 0.001$). There was a significant effect of time ($p = 0.003$) for RE, being higher at M4 and M5 compared with M1 and M3. Significant correlations were found between marathon performance and speeds at AeT and AnT at every time point ($r = 0.81$ – 0.94 ; $p < 0.05$). Speed at AnT represented the main influence (65.9 and 71.41%) in the final time prediction at M1 and M2, whereas speed at AeT took its place toward the end of the macrocycle (76.0, 80.4, and 85.0% for M3, M4, and M5, respectively). In conclusion, assessment of speeds at AeT and AnT permits for reasonable performance prediction during the training preparation, therefore avoiding maximal testing while monitoring 2 fundamental training speeds. Future research should verify if these findings are applicable to runners of different levels and other periodization models.

Key Words: running performance prediction, endurance training, anaerobic threshold, running economy

Introduction

Marathons have become the most popular endurance event with millions of recreational athletes participating each year in competitions worldwide. Marathon represents the ultimate challenge for many runners, and therefore, they must prepare themselves both physically and mentally for such a strenuous event (3,33). In this scenario, improving an athlete's physical and physiological capacities is one of the biggest puzzles to be addressed by athletes, coaches, and sport physiologists. This, in time, implies the knowledge of those factors associated with successful performance in the desired athletic event. As an example, there is enough evidence to support the notion that maximal oxygen uptake ($\dot{V}O_{2\max}$) is one of the most important physiological factors contributing to endurance running performance (2,16,22,24,26). However, $\dot{V}O_{2\max}$ does not solely explain endurance running performance, particularly in long-distance events such the marathon. For instance, although high values of $\dot{V}O_{2\max}$ could be important for performance time, long-distance runners usually have lower values when compared with shorter distance athletes because other factors such as fractional utilization of (%) $\dot{V}O_{2\max}$ or the highest sustainable % $\dot{V}O_{2\max}$ and

running economy (RE) become more important for performance time in long-distance running (16,17,26,29).

To improve running performance, it is essential to establish which of those factors contributing to competitive performance are to be enhanced by training. To accomplish this goal, it is required that performance could be predicted because, from a practical point of view, predicting competitive time will retrieve useful information for coaches and athletes allowing them to better prepare for both training (e.g., tempo runs) and competition (e.g., pacing strategies). Moreover, predicting running performance helps to understand the factors associated with the ability to sustain a given speed for a given exercise duration leading to a successful performance time. In this sense, several approaches have been used to predict running performance time. For instance, expected times based on shorter distances are commonly used to estimate an athlete's potential during longer events (21). Particularly, nomograms are an appealing method for predicting performance time (5). However, performance time prediction using nomograms is based on the assumption of equivalence between performances in different events. Therefore, although its simplicity and validity can be attractive for coaches and athletes, given that it requires the performance times of 2 distances and marathons and shorter races are usually run few times on a year, this assumption is not always met. In addition, the accuracy in the predictions seems to be higher when performance is obtained by interpolation instead of extrapolation (5).

Address correspondence to Jonathan Esteve-Lanao, jonathan.esteve@allinyourmind.es.

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Other approaches for predicting marathon performance time includes the use of physiological variables. Since the mid-70s, researchers have tried to predict performance times using different physiological testing variables (11,30). For instance, Riegel (27) intended to predict endurance performance times (including the marathon) using an “endurance equation” that includes a fatigue constant to describe time, speed, and distance. Later models have fitted several variables into equations, such as those obtained during laboratory testing (e.g., $\dot{V}O_2\text{max}$, anaerobic threshold [AnT]), anthropometric measures, and training loads (i.e., intensity, duration, and frequency of training), to predict the mean speed of a marathon race (1,6,10,14,15,17,24,28,32). As AnT correlates with aerobic capacity in marathon runners (25), it is often used for training prescription because it serves to determine different speed zones (4,7,31). In addition, AnT combined with RE can better predict long-distance running performance than AnT alone (17). Thus, tracking and quantifying the changes of these parameters during the training season could be very helpful for both training prescription and performance time prediction. However, to the best of our knowledge, no previous studies have addressed this. Therefore, the objective of this study was to predict marathon final speed from different moments along the training season, considering different speed zones, derived from AnT, and RE as independent variables.

Methods

Experimental Approach to the Problem

A group of recreational runners performed a 16-week macrocycle before a marathon and were evaluated in a laboratory at different time points for determining the best physiological predictors of performance time during different phases of the preparation.

Subjects

Sixteen recreational runners (8 women and 8 men), who were to compete in a marathon race on the same date, participated in the study. All runners shared the same coach who directly supervised each training session. A minimum sample size of 13 runners was calculated using G-Power software (8) assuming a 0.05 significance level, a 0.80 power, and a correlation ≥ 0.70 among the 5 repeated

measures, to detect medium effect sizes. We increased the required sample size up to 16, to buffer against attrition. Eventually, the initial sample completed the entire training program. The inclusion criteria were as follows: (a) must have been training for endurance running events (i.e., marathons and half-marathons) during at least the past 1.5 years, (b) must have competed in, at least, one marathon race, and (c) must have represented, as sex groups, 2 homogeneous samples in terms of relative performance (vs. marathon race winners). The exclusion criteria were as follows: (a) the presence of any kind of cardiovascular disease, (b) must have missed more than 3 consecutive sessions by cause of an injury or disease, and (c) must have accomplished less than 90% of the training sessions as specified at the program. Before engaging in any test, subjects were informed about the risks and benefits of the study and provided written informed consent. The protocol of the study was approved by the Ethics Committee of the European University of Madrid (CIPI/035/15). Descriptive characteristics of the runners are shown in Table 1.

Procedures

Training Program. All runners completed a 16-week macrocycle following a reverse periodization model. The training program was composed of 4 mesocycles (4 weeks for each mesocycle) with a 3:1 structure; i.e., the training load increased linearly during the first 3 microcycles, and this was followed by a week of reduced load to facilitate recovery and supercompensation. The training intensity was set to be mainly at the ventilatory threshold (VT) during the initial weeks and moving progressively toward a “between thresholds—marathon pace” intensity. On the other hand, the progression of volume was set to be from low to high along the macrocycle, achieving its peak at week 11 and with the longest distance ran at the end of week 13 (21 days before the marathon race). Before the specific training period, these runners followed 3–6 months of regular training and focused on 10 and 21 k competition distances. During the previous 6 weeks before the specific program, the runners followed a very similar training program considered the general preparation period, which served as the general adaptation before the study. At the end of this period (week 0), the physiological testing battery M1 was conducted as described below (Table 2).

Table 1
Sample characteristics before the study ($n = 16$, 8 men and 8 women).*

	All Mean $\pm$ SD (CI 95%)	Women Mean $\pm$ SD (CI 95%)	Men Mean $\pm$ SD (CI 95%)	<i>p</i>	CI 95% for the difference
Age (y)	36.9 $\pm$ 4.7 (34.4 to 39.4)	36.3 $\pm$ 5.6 (31.6 to 4.9)	37.6 $\pm$ 3.9 (34.4 to 40.9)	0.576	−3.78 to 6.53
Mass (kg)	66.8 $\pm$ 14.9 (58.4 to 74.7)	55.1 $\pm$ 4.5 (51.3 to 58.6)	78.5 $\pm$ 11.8 (68.7 to 88.4)	0.001	13.85 to 33.02
Height (m)	1.67 $\pm$ 0.1 (1.61 to 1.72)	1.60 $\pm$ 0.06 (1.55 to 1.65)	1.73 $\pm$ 0.09 (1.66 to 1.81)	0.003	0.05 to 0.21
BMI ($\text{kg}\cdot\text{m}^{-2}$)	23.8 $\pm$ 3.0 (22.1 to 25.3)	21.5 $\pm$ 1.8 (16.9 to 23.0)	26.0 $\pm$ 2.2 (24.2 to 27.8)	0.001	2.39 to 6.70
Endurance training experience (y)	2.9 $\pm$ 1.1 (2.5 to 4.1)	3.0 $\pm$ 1.4 (2.2 to 5.2)	2.8 $\pm$ 0.9 (1.9 to 3.8)	0.255	−2.45 to 0.71
Study's marathon performance (hh:mm)	04 h 13 min $\pm$ 38 min	04 h 17 min $\pm$ 44 min	04 h 08 min $\pm$ 32 min	0.668	
Study's marathon performance (% sex's absolute winner time)	189 $\pm$ 29 (173 to 204)	181 $\pm$ 29 (156 to 205)	197 $\pm$ 27 (174 to 219)	0.269	−14.03 to 46.54
Mean speed difference between first and second half-marathon (race d)	−2.6 $\pm$ 4.7 (−5.0 to −0.1)	−2.4 $\pm$ 3.9 (−5.6 to 0.9)	−2.8 $\pm$ 5.6 (−7.4 to 1.9)	0.871	−5.57 to 4.77
HR ($\text{b}\cdot\text{min}^{-1}$) at VT (zone 2)	158 $\pm$ 10 (153 to 163)	157 $\pm$ 11 (148 to 166)	159 $\pm$ 9 (151 to 167)	0.716	−8.96 to 12.71
HR ($\text{b}\cdot\text{min}^{-1}$) at RCT (zone 4)	176 $\pm$ 9 (171 to 180)	177 $\pm$ 9 (170 to 184)	175 $\pm$ 9 (166 to 182)	0.550	−12.39 to 6.89
HRmax ($\text{b}\cdot\text{min}^{-1}$)	184 $\pm$ 9 (179 to 189)	184 $\pm$ 7 (175 to 193)	185 $\pm$ 11 (177 to 189)	0.810	−10.96 to 8.71
$\dot{V}O_2\text{max}$ ($\text{ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$)	48.0 $\pm$ 6.4 (44.6 to 51.3)	45.2 $\pm$ 4.3 (41.5 to 48.8)	50.8 $\pm$ 7.1 (44.8 to 56.7)	0.077	−0.69 to 11.87
Velocity at $\dot{V}O_2\text{max}$ ($\text{km}\cdot\text{h}^{-1}$)	14.2 $\pm$ 1.3 (13.4 to 14.8)	13.6 $\pm$ 1.0 (12.7 to 14.5)	14.7 $\pm$ 1.4 (13.5 to 15.8)	0.107	−0.25 to 2.38
RE testing velocity (% mean marathon speed)	104 $\pm$ 7 (100 to 108)	105 $\pm$ 9 (98.2 to 112.8)	104 $\pm$ 6 (98.8 to 108.4)	0.618	−9.76 to 6.01

*CI = confidence intervals; BMI = body mass index; HR = heart rate; $\text{b}\cdot\text{min}^{-1}$ = beats per minute; VT = ventilatory threshold (first metabolic threshold); RCT = respiratory compensation threshold (second metabolic threshold); RE = running economy.

Table 2**Training program general schedule.***

Period	General						Specific															
	-2			-1			1				2				3				4			
	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Mesocycle																						
Week #	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Weeks to race	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
Days to race	153	146	139	132	125	118	111	104	97	90	83	76	69	62	55	48	41	34	27	20	13	6
Testing #						M1				M2				M3				M4			M5	

*Weeks/days to race are count from the beginning of the week to the race day. Race day was Sunday. Tests were conducted as described in the Methods section.

Physiological Testing: Maximal Graded Test. At the beginning of the study, a maximal graded running test was conducted to collect physiological data. The test was conducted on a motorized treadmill (Full Vision TMX425; Full Vision Drive, Newton, KS, USA) with a 1% grade and started at 8.0 km·h⁻¹ with increases of 0.3 km·h⁻¹ every 30 seconds until volitional exhaustion. This protocol has been previously demonstrated to be appropriate for recreational runners' evaluation (23). During the test, a breath-by-breath system was used for gas exchange data collection (Med Graphics PFX Ultima; Medical Graphics Corporation, St Paul, MN, USA), and the heart rate (HR) was continuously monitored by the integrated electrocardiogram system (CardioPerfect PRO; Welch Allyn Cardio Control BV, Delft, Netherlands) and a HR monitor (Polar RC3 Global Positioning System (GPS); Polar Electro Oy, Kempele, Finland). Gas exchange data were used to determine maximal oxygen uptake ($\dot{V}O_{2max}$) and metabolic thresholds (VT or aerobic threshold; respiratory compensation threshold [RCT]) using previously described methods (23). Also, maximal HR (HR_{max}), peak speed (Speak), and pace and HR at the metabolic thresholds (HR at VT and HR at RCT) were computed. This initial test was conducted only once, assuming that, in endurance-trained athletes, the thresholds' associated HR determined during laboratory testing remains stable over the season despite significant improvements in the workload eliciting both thresholds (20). All these parameters have previously demonstrated excellent reliability values (i.e., intraclass correlation coefficients >0.9).

Running Economy. For RE determination, runners completed a constant load treadmill running test of 15 minutes duration at a 1% grade, before the maximal graded test during the first testing session. All subjects avoided hard training sessions 3 days before evaluations on an individual basis. The speed for the test was selected according to the individual's expected marathon pace. Once this speed was decided and used for the first testing session, it was kept constant through the remaining tests (second to fifth). Breath-by-breath $\dot{V}O_2$ was measured and averaged every 30 seconds during the test. Steady state was defined as an increase of

100 ml of O₂ over the final 2 minutes of each stage. Since Fletcher et al. (9) suggested that RE calculations might be different in case of including respiratory quotient or considering O₂, RE was expressed as both a gross oxygen cost (ml O₂·kg⁻¹·km⁻¹) and gross caloric unit cost (kcal·kg⁻¹·km⁻¹). All laboratory tests were conducted on the same treadmill (Full Vision TMX425; Full Vision Drive), and both temperature (20° C) and humidity (65%) conditions were kept constant. A fan was placed 50 cm in front of the runners, at their chest height, to limit sweating discomfort and to allow for refrigeration (19). All these parameters have previously demonstrated excellent reliability values (i.e., intraclass correlation coefficients >0.9).

Testing and Training Monitoring Along the Training Macrocycle. Based on the physiological measures described in previous sections, 6 aerobic training zones were set for training purposes (Table 3). The so-called "zone 2" corresponded to the VT intensity (AeT), whereas the "zone 4" corresponds to the RCT intensity (AnT). For training purposes, intensity was based on the HR associated with speeds at the different zones. Thus, each runner wore a HR monitor (Polar RC3 GPS; Polar Electro Oy) for monitoring these intensities in each training session. Also, to monitor and assess the training effect on RE and speeds at zones 2 and 4, 2 physiological tests were conducted at 5 different time points (M1, M2, M3, M4, and M5) along the training macrocycle (Table 2). The testing schedule was always the same on a weekly basis. Two days were interspersed between the last "hard" (long-distance) running session and the assessment. One "easy" day was placed between both tests (RE and zones 2/4 speed).

Speed at Ventilatory Threshold and Respiratory Compensation Threshold. The speeds eliciting zone 2 and zone 4 were averaged from two 20 minutes bouts conducted on a synthetic 400-m track interspersed by a 5-minute passive rest. Runners were instructed to keep a continuous speed, based on the initial graded test at M1 and in recent training workouts at these zones. They were specifically instructed to reach the target HR zone by minute 3 and to

Table 3**Individual training zones identified before the study.*†**

Zone	Criteria	Example (subject #12)	
		HR zones (b·min ⁻¹)	Pace zones (min:s range per km)
1	15 to 1 b·min ⁻¹ below VT	142–157	6:30–5:53
2	From VT HR to +3 b·min ⁻¹	158–161	5:52–5:33
3	Between zones 2 and 4	162–175	5:32–5:01
4	RCT HR to +3 b·min ⁻¹	176–179	5:00–4:53
5	Between zones 4 and 6	180–184	4:52–4:33
6	<3 b·min ⁻¹ to max HR	185–188	4:32–4:25

*HR = heart rate; VT = ventilatory threshold; RCT = respiratory compensation threshold.

†See details in the text for training and testing purposes.

stay in the HR zone for the remaining time of each bout. GPS-system watches were allowed, but the speed measures taken were those from the 400-m track laps, to get comparable measurements. The average time of fully covered laps was considered to determine the speed of a given metabolic zone (i.e., 400-m portions were excluded in case of reaching the 20 minutes before the end of a lap). Thus, the average speed of at least ~18-minute bouts was considered. All these parameters have previously demonstrated excellent reliability values (i.e., intraclass correlation coefficients >0.9).

All measurements were conducted early in the morning (5:00–6:30 AM) under similar weather conditions (no rain, no wind, temperature 23–25° C, humidity 70–75%), to avoid the extreme environmental conditions of the tropical climate in Yucatan during subsequent hours, and because it is the same time for training sessions and competitions. The training workout and the RE session were both preceded by a 15-minute easy jog at training zone 1, followed by a dynamic stretching workout, in a regular fashion.

Marathon Timing. A shoe-laced electronic timing system provided by the organizers was used to record personal times. Partial times were provided so that it was calculated the pace drop between the 2 halves of the race.

Statistical Analyses

Descriptive data are presented as mean $\pm$ SD and 95% confidence intervals. Normality was assessed by means of standard statistical parameters (i.e., skewness and kurtosis), graphical methods (Q-Q plots and box plots), and using the Shapiro–Wilk normality test. A repeated-measures analysis of variance (ANOVA) with 5 levels (M1 vs. M2 vs. M3 vs. M4 vs. M5) was used to test for differences in speeds at AnT and AeT and for RE along the training macrocycle. Also, a 2 (men vs. women) $\times$ 5 (M1 vs. M2 vs. M3 vs. M4 vs. M5) mixed ANOVA was applied to test for differences in AnT, AeT, and RE between sexes along the macrocycle. Homogeneity of variance was checked using the Levene test, and sphericity was controlled using the Mauchly test. If sphericity could not be met, then the Greenhouse–Geisser

correction was used instead. Effect sizes were calculated using η^2 and Cohen's d . As the main objective of this study was to evaluate whether speeds at the metabolic thresholds (i.e., AeT and AnT) assessed at different time points could predict marathon speed, a hierarchical linear regression was conducted using the aforementioned speeds as main predictors. In addition, within each regression, several variables (i.e., RE, $\dot{V}O_{2\max}$, Speak, HRmax, years of experience, body mass index, etc.) were added to the model to investigate whether adding those variables would improve performance prediction. For a regression, to be regarded as a valid prediction of marathon time, the following assumptions were tested: (a) independence of observations (i.e., independence of residuals), (b) linear relationship between the dependent variable and each of the independent variables, and between the dependent variable and the independent variables collectively, (c) homoscedasticity of residuals (equal error variances), (d) multicollinearity, (e) significance of outliers, and (f) normal distribution of errors. If one or more of these assumptions were violated, then several corrections were applied in an attempt to solve the problem. Each regression was tested for the sample as a whole and independently for men and women to control for potential sex differences. Adjusted R^2 was used to assess the percentage of variance explained by linear regression functions. Only large correlations (≥ 0.50) were considered statistically significant, according with the aims of the study. Simple correlations were performed by means of Pearson product-moment correlations. All statistics were performed with the IBM-SPSS software for Windows (Version 20.0; Armonk, NY). The statistical significance was set at an alpha level of 0.05.

Results

As shown in Table 4, when the whole sample was considered, there was a significant effect of time, both speed at AeT [$F_{(2,07-31,14)} = 9.136$; $p = 0.001$, $\eta^2 = 0.38$] and AnT [$F_{(5-75)} = 4.957$; $p = 0.001$, $\eta^2 = 0.25$]. Paired comparisons showed that both speed at AeT and AnT improved at M2 vs. M1 and remained significantly higher than M1 at M3, M4, and M5. Also, for RE expressed in $\text{mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ there was significant

Table 4
Speeds at zone 2 (AeT) and zone 4 (AnT) thresholds and running economy (RE) along the training macrocycle for the whole sample.*†

	M1	M2	M3	M4	M5	CI 95% for the difference	ES
Speed at AeT ($\text{km}\cdot\text{h}^{-1}$)	10.57 $\pm$ 1.36	10.88 $\pm$ 1.24‡	11.01 $\pm$ 1.18‡	11.22 $\pm$ 1.21‡	11.31 $\pm$ 1.26‡	M2 vs. M1 (−0.47 to −0.15) M3 vs. M1 (−0.65 to −0.23) M4 vs. M1 (−0.87 to −0.44) M5 vs. M1 (−1.06 to −0.42)	0.23 0.34 0.50 0.55
Speed at AnT ($\text{km}\cdot\text{h}^{-1}$)	12.00 $\pm$ 1.46	12.36 $\pm$ 1.35‡	12.56 $\pm$ 1.50‡	12.54 $\pm$ 1.38‡	12.56 $\pm$ 1.48‡	M2 vs. M1 (−0.60 to −0.13) M3 vs. M1 (−0.82 to −0.28) M4 vs. M1 (−0.79 to −0.28) M5 vs. M1 (−0.87 to −0.24)	0.25 0.37 0.37 0.37
RE ($\text{mL}\cdot\text{kg}^{-1}\cdot\text{km}^{-1}$)	216.38 $\pm$ 21.49	209.31 $\pm$ 15.10	212 $\pm$ 12.69	205.56 $\pm$ 16.80‡	201.94 $\pm$ 13.08‡§	M4 vs. M1 (1.10 to 20.51) M5 vs. M1 (4.25 to 24.61) M5 vs. M2 (0.76 to 13.98) M4 vs. M3 (0.32 to 12.67) M5 vs. M3 (3.48 to 16.76)	0.55 0.79 0.51 0.43 0.77
RE ($\text{kcal}\cdot\text{kg}^{-1}\cdot\text{km}^{-1}$)	1.08 $\pm$ 0.10	1.05 $\pm$ 0.7	1.06 $\pm$ 0.6	1.05 $\pm$ 0.9	1.03 $\pm$ 0.6*	M5 vs. M1 (0.01 to 0.09) M5 vs. M3 (0.01 to 0.06)	0.61 0.52

*AeT = aerobic threshold; AnT = anaerobic threshold; CI 95% = confidence intervals for every significant difference, ES = effect size (Cohen's d).

†Values are mean $\pm$ SD and 95%.

‡Significantly different from M1 ($p < 0.05$).

§Significantly different from M2 ($p < 0.05$).

||Significantly different from M3 ($p < 0.05$).

effect of time [$F_{(4-60)} = 4.957$, $p = 0.003$, $\eta^2 = 0.23$] being higher at M4 and M5 compared with M1 and M3 (Table 4), whereas RE expressed in $\text{kcal}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ was only higher at M5 vs. M1 and M3.

When analyses were performed by sex, there was neither a significant main simple effect of sex nor a time $\times$ sex interaction. However, there were significant main simple effects of time for speed at AeT [$F_{(2.04-28.62)} = 13.967$; $p < 0.001$, $\eta^2 = 0.49$] and speed at AnT [$F_{(4-56)} = 7.342$; $p < 0.001$, $\eta^2 = 0.35$]. Interestingly, post hoc analyses revealed that speed at AeT increased from M1 to M2 and remained elevated in M3, M4, and M5 compared with M1, with a further increase in M4 compared with M2 and M3 for men, and in M5 compared with M3 and M4 for women (Table 5). On the other hand, speed at AnT only showed a significant effect of time for men (Table 5). There were no significant differences for RE expressed both as $\text{mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ and $\text{kcal}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$.

Significant correlations were found between marathon speed and predictor speeds (speed at AeT and speed at AnT) at every time point ($r = 0.81-0.94$; $p < 0.05$). However, none of the economy measurements (RE_{mL} or RE_{kcal}) showed statistically significant correlation with marathon speed. We calculated the best predictive equations for every moment (M1–M5) using only the speeds at AeT and AnT. Using

Equation 1, we predicted 65.9% of marathon speed with speed at AnT; using Equation 2, we predicted 71.4% with speed at AnT; with Equations 3–5, speed at AeT values were used to predict 76.0, 80.4, and 85.0% of marathon speed, respectively. Adding other variables to the models did not result in better performance predictions.

$$116 \text{ days : Avg42k speed (km}\cdot\text{h}^{-1}) = \text{speed at AnT (km}\cdot\text{h}^{-1}) \times 0.771 + 0.959 \text{ (CI 95\% [0.47–1.07])}. \quad (1)$$

$$88 \text{ days : Avg42k speed (km}\cdot\text{h}^{-1}) = \text{speed at AnT (km}\cdot\text{h}^{-1}) \times 0.863 + 1.463 \text{ (CI 95\% [0.56–1.16])}. \quad (2)$$

$$60 \text{ days : Avg42k speed (km}\cdot\text{h}^{-1}) = \text{speed at AeT (km}\cdot\text{h}^{-1}) \times 1.013 + 0.944 \text{ (CI 95\% [0.70–1.32])}. \quad (3)$$

$$32 \text{ days : Avg42k speed (km}\cdot\text{h}^{-1}) = \text{speed at AeT (km}\cdot\text{h}^{-1}) \times 1.012 + 1.147 \text{ (CI 95\% [0.73–1.28])}. \quad (4)$$

$$11 \text{ days : Avg42k speed (km}\cdot\text{h}^{-1}) = \text{speed at AeT (km}\cdot\text{h}^{-1}) \times 1.004 + 1.145 \text{ (CI 95\% [0.77–1.23])}. \quad (5)$$

Table 5
Speeds at zone 2 (AeT) and zone 4 (AnT) thresholds and running economy (RE) along the training macrocycle by sex.*†

	Males					CI 95% for the difference	ES
	M1	M2	M3	M4	M5		
Speed at AeT ($\text{km}\cdot\text{h}^{-1}$)	10.86 $\pm$ 1.27	11.11 $\pm$ 1.24‡	11.29 $\pm$ 1.09‡	11.55 $\pm$ 1.09‡§	11.43 $\pm$ 1.13‡	M2 vs. M1 (–0.46 to –0.05) M3 vs. M1 (–0.76 to –0.09) M4 vs. M1 (–1.02 to –0.35) M5 vs. M1 (–1.04 to –1.07) M4 vs. M2 (–0.80 to –0.06) M4 vs. M3 (–0.52 to –0.01)	0.19 0.34 0.55 0.45 0.35 0.23
Speed at AnT ($\text{km}\cdot\text{h}^{-1}$)	12.23 $\pm$ 1.30	12.47 $\pm$ 1.09	12.94 $\pm$ 1.34‡§	12.87 $\pm$ 1.17‡	12.97 $\pm$ 1.15‡§	M3 vs. M1 (–1.16 to –0.26) M4 vs. M1 (–1.05 to –0.23) M5 vs. M1 (–1.11 to –0.38) M3 vs. M2 (–0.90 to –0.02) M5 vs. M2 (–0.82 to –0.18)	0.51 0.49 0.58 0.36 0.42
RE ($\text{mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$)	219.75 $\pm$ 18.62	213.75 $\pm$ 13.65	216.38 $\pm$ 14.92	209.75 $\pm$ 19.43	207.00 $\pm$ 14.55		
RE ($\text{kcal}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$)	1.10 $\pm$ 0.09	1.06 $\pm$ 0.06	1.08 $\pm$ 0.07	1.07 $\pm$ 0.10	1.05 $\pm$ 0.08		
	Females					CI 95% for the difference	ES
	M1	M2	M3	M4	M5		
Speed at AeT ($\text{km}\cdot\text{h}^{-1}$)	10.28 $\pm$ 1.47	10.65 $\pm$ 1.28‡	10.74 $\pm$ 1.28‡	10.90 $\pm$ 1.32‡	11.20 $\pm$ 1.44‡ ¶	M2 vs. M1 (–0.66 to –0.06) M3 vs. M1 (–0.80 to –0.10) M4 vs. M1 (–0.98 to –0.25) M5 vs. M1 (–1.42 to –0.41) M5 vs. M3 (–0.84 to –0.07) M5 vs. M4 (–0.56 to –0.02)	0.25 0.31 0.42 0.59 0.32 0.20
Speed at AnT ($\text{km}\cdot\text{h}^{-1}$)	11.77 $\pm$ 1.66	12.26 $\pm$ 1.64	12.17 $\pm$ 1.65	12.21 $\pm$ 1.56	12.15 $\pm$ 1.72		
RE ($\text{mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$)	213.00 $\pm$ 24.84	204.88 $\pm$ 16.04	207.75 $\pm$ 8.94	201.38 $\pm$ 13.68	196.88 $\pm$ 9.80		
RE ($\text{kcal}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$)	1.06 $\pm$ 0.11	1.04 $\pm$ 0.08	1.04 $\pm$ 0.03	1.02 $\pm$ 0.07	1.01 $\pm$ 0.05		

*AeT = aerobic threshold; AnT = anaerobic threshold.

†Values are mean $\pm$ SD, confidence intervals (CI 95%), and effect sizes (ES, Cohen's d) for the differences.

‡Significantly different than M1 ($p > 0.05$).

§Significantly different from M2 ($p < 0.05$).

||Significantly different from M3 ($p < 0.05$).

¶Significantly different from M4 ($p < 0.05$).

Discussion

The main objective of this study was to evaluate the power for marathon performance prediction of speeds at 2 training zones (AeT and AnT) and the change in RE during a training macrocycle. To the best of our knowledge, we have shown for the first time that training speed zones, specifically those associated with AeT and AnT, can be used to predict marathon running, in recreational runners, at different time points during a training macrocycle. Surprisingly, despite the significant improvements in RE toward the end of the training program (Table 4), this factor was not able to predict marathon running speed.

Previous research has shown that long-distance running performance requires high values of $\dot{V}O_{2\max}$ and sustainable $\% \dot{V}O_{2\max}$ together with good RE (16,17,26,29). However, at similar values of $\dot{V}O_{2\max}$, anaerobic threshold and RE are likely to be determinant for performance (18,19). In our study, it becomes particularly interesting that RE would not fit into any model, no matter what index was used. Nevertheless, from a conceptual point of view, RE is already included when observing a speed developed at a given zone. In this sense, as previously shown, the speeds associated with physiological variables have been the strongest predictors (17,24,28). However, when analyzed alone, RE showed a significant, nonlinear improvement along the specific period, reaching the highest level before competition. The reason why it was not acting as a pure determinant factor, as classically suggested for this event (17), remains unclear, but may be related to the sample's heterogeneity. Initial RE (M1) was between 182 and 197 $\text{ml}\cdot\text{kg}^{-1}\cdot\text{km}^{-1}$ in 3 of our runners. These values are close to what has been determined in top elite runners (19). These “most economical runners” did not show great (if any) improvements during the program. On the other hand, the RE was between 241 and 262 $\text{ml}\cdot\text{kg}^{-1}\cdot\text{km}^{-1}$ in other 3 runners, which represents a very poor level (12). As a group, a significant improvement was found at the end of the program. Thus, it can be suspected that a “ceiling effect” was reached for those most economical runners, whereas others improved a lot. Less heterogeneity and bigger sample size might be controlled in future studies to obtain clearer view of the predictive value of RE for marathon performance.

Recently, Gomez-Molina et al. (13), using a stepwise multiple regression analyses, were able to predict half-marathon performance with the inclusion of training, anthropometrical, physiological, and biomechanical variables. Interestingly, the highest estimation power (96.2%) was obtained when Speak and speed at RCT together with running experience were included in the model. Taken into consideration previous and current results, it could be suggested that speed at thresholds is not only among the main determinants of performance but also has a high predictive value. However, given that half-marathon is run at a faster pace compared with marathon, the additive power of Speak for predicting half-marathon performance would seem logical.

The conducted regression analyses (i.e., the equations found) can be meaningful from an applied point of view. Overall, for the whole sample it was found that speed at AnT (zone 4) represented the main influence in the final time prediction at M1 and M2, whereas speed at AeT (zone 2) took its place toward the end of the training macrocycle (M3, M4, and M5). A direct application of this study was to identify a precise moment to establish accurate predictions on marathon final time. The significant correlations

found showed gradually higher r coefficients, meaning that the accuracy of the marathon time prediction was higher. Thus, coaches should consider whether it is proper or not to set a wider range so far from the competition or awaiting a higher accuracy despite the growing levels of anxiety by a lack of premature prediction. Also, coaches could apply these regression models for monitoring training and thus set training goals accordingly to an athlete's expectancies.

Finally, it is worth noting that other prediction methods (e.g., expected times based on shorter distances) are commonly used to estimate an athlete's potential during longer events and can also be useful for coaches and athletes (21). However, performance prediction requires the performance of 2 distances, and marathons and shorter races are usually run few times on a year. Hence, given that metabolic thresholds have the highest predictive power for marathon performance (29), our approach would permit for reasonable performance prediction while monitoring, at the same time, the changes in 2 fundamental training variables.

For recreational athletes, marathon time is better predicted by zone 4 speed (i.e., AnT) up to 13 weeks before the race, whereas zone 2 speed (i.e., AeT) better predicts marathon time with less of 9 weeks until the race. In addition, the accuracy of the proposed equations is higher as the date of marathon is closer. Other parameters such as RE or Speak did not improve the predictive models proposed. Future research should verify whether these findings could be extended to runners of different levels and other periodization models.

Practical Applications

From the current results, it could be recommended to only evaluate physiological thresholds as AnT and AeT during the preparation of a marathon to determine training speed zones and predict running performances in recreational athletes of both sexes. This means that maximal testing could be avoided because other parameters as Speak did not improve our predictive models. Coaches should decide how to manage the proper moment and number of assessments to cope with runners anxiety about predictions, while establishing individual training goals and racing strategies.

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